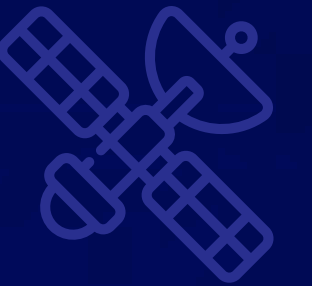
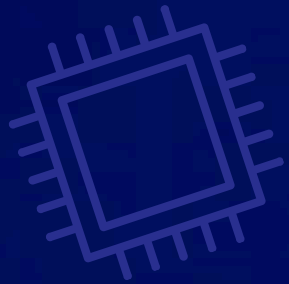
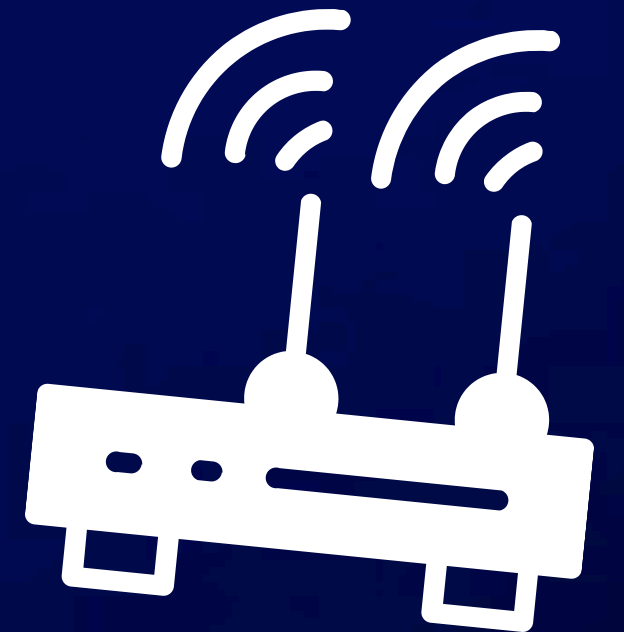
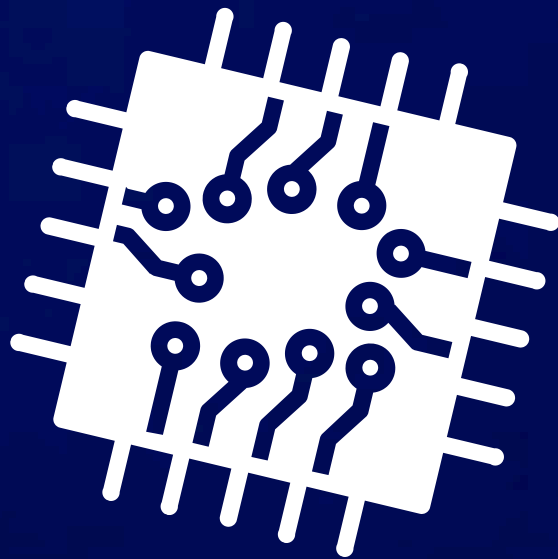


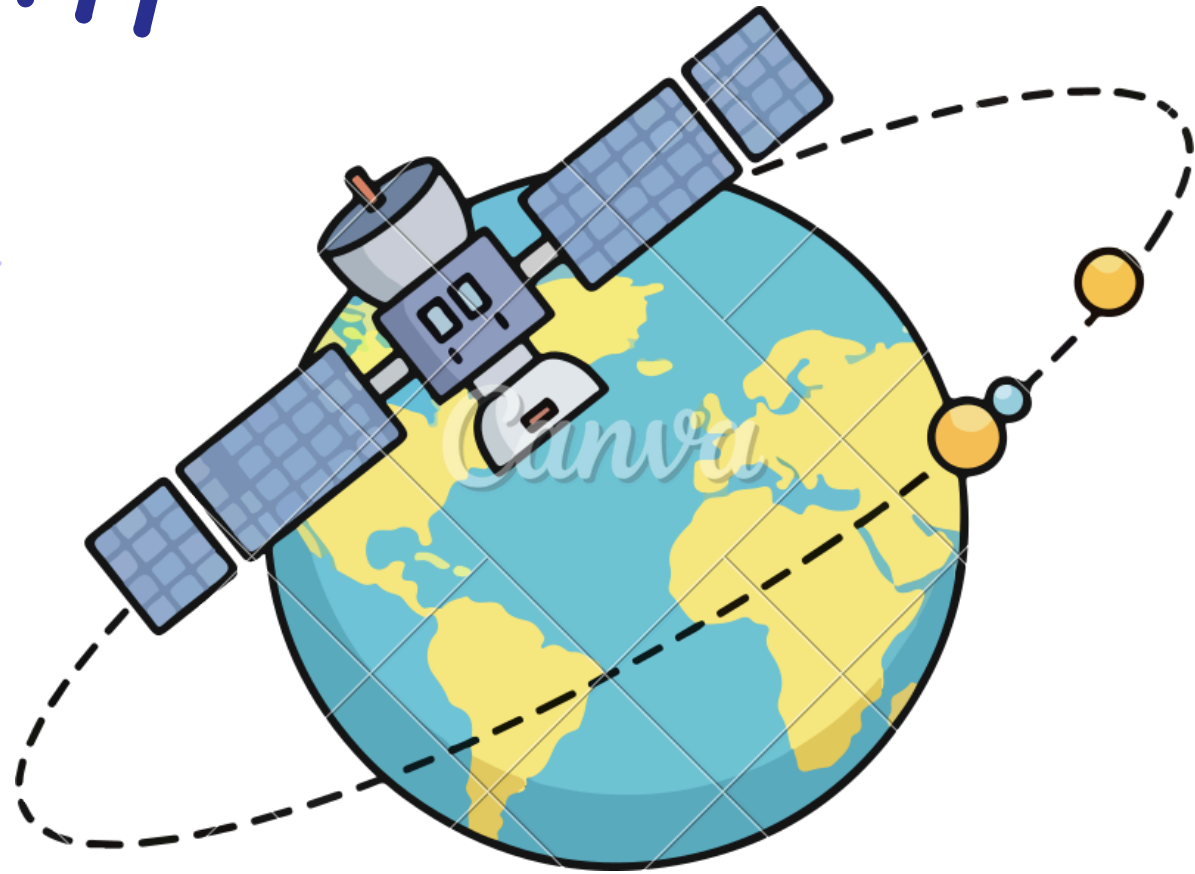
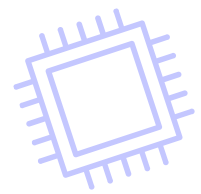
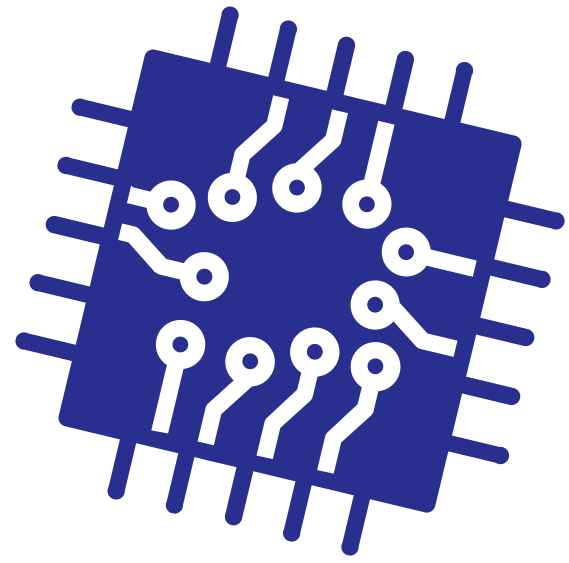


GNSS_R SYSTEM



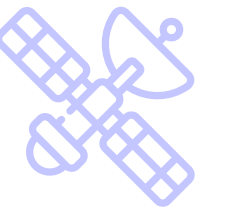
Presented by Erick Tchitae





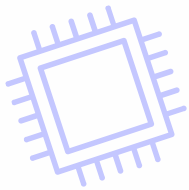
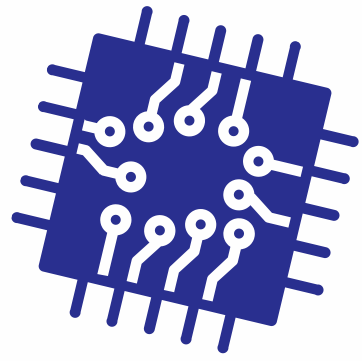
Summary

- 01 Introduction
- 02 Literature review
- 03 System Architecture
- 04 Integration of the Mobile and Static Units
- 05 Result
- 06 Conclusion





INTRODUCTION



GPS

- Owned and operated by the United States of America
- First satellite launched in 1978
- Available globally



GLONASS

- Owned and operated by Russia
- First satellite launched in 1982
- Available globally



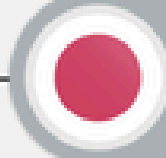
BeiDou

- Owned and operated by China
- First satellite launched in 2000
- Available globally



Galileo

- Owned and operated by the European Union (E.U.)
- First satellite launched in 2011
- Available globally



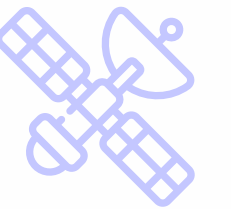
QZSS

- Owned and operated by Japan
- First satellite launched in 2010
- Available regionally



IRNSS

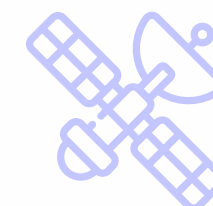
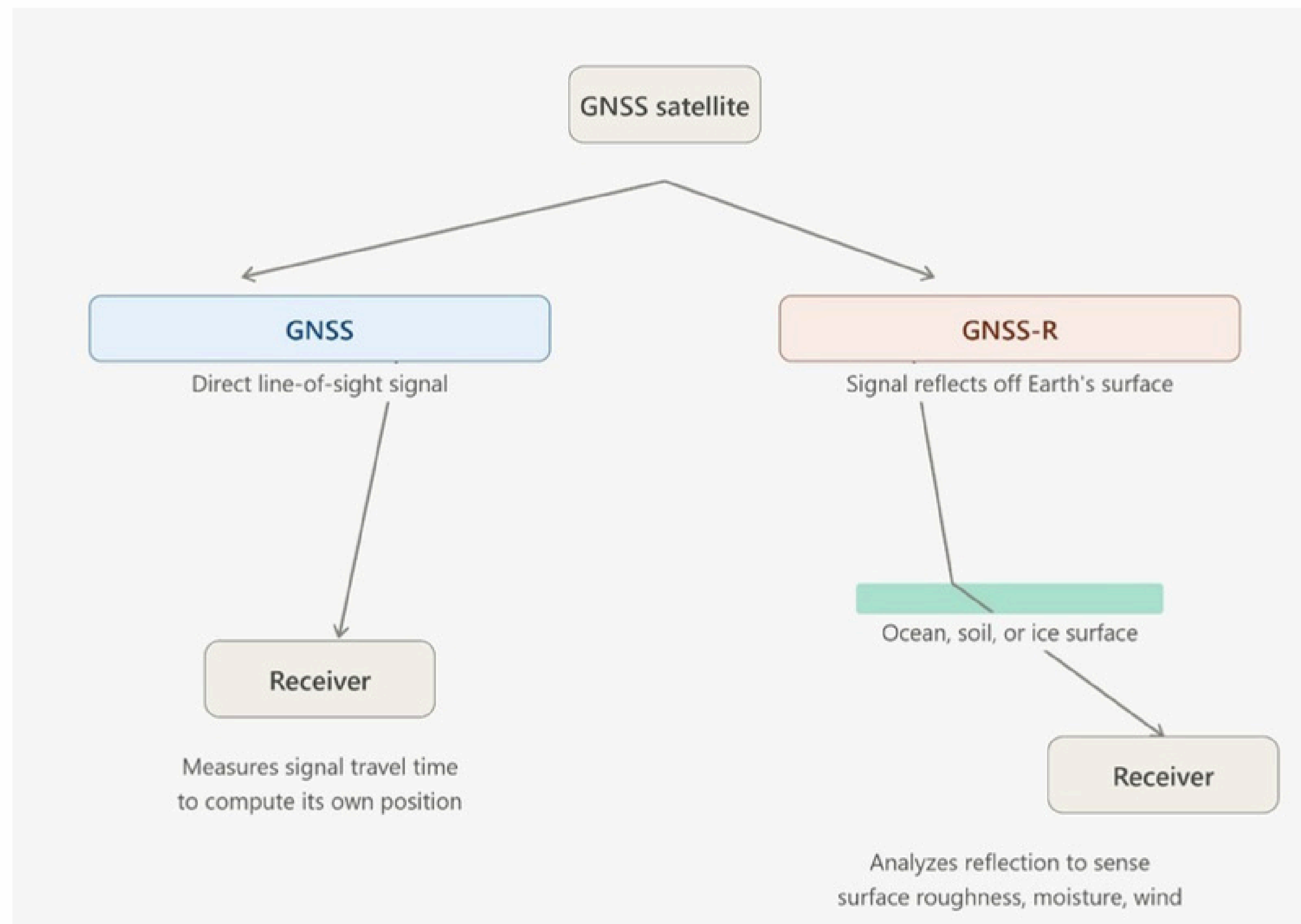
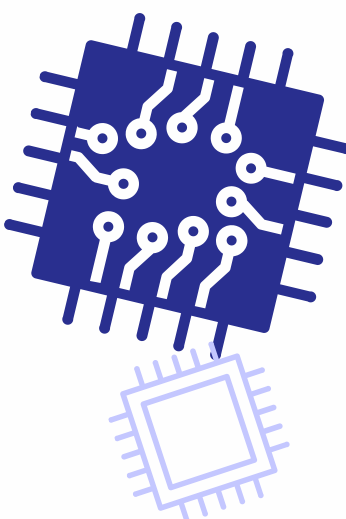
- Owned and operated by India
- First satellite launched in 2013
- Available regionally

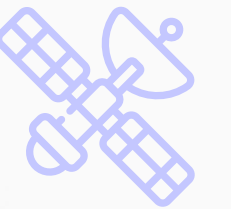


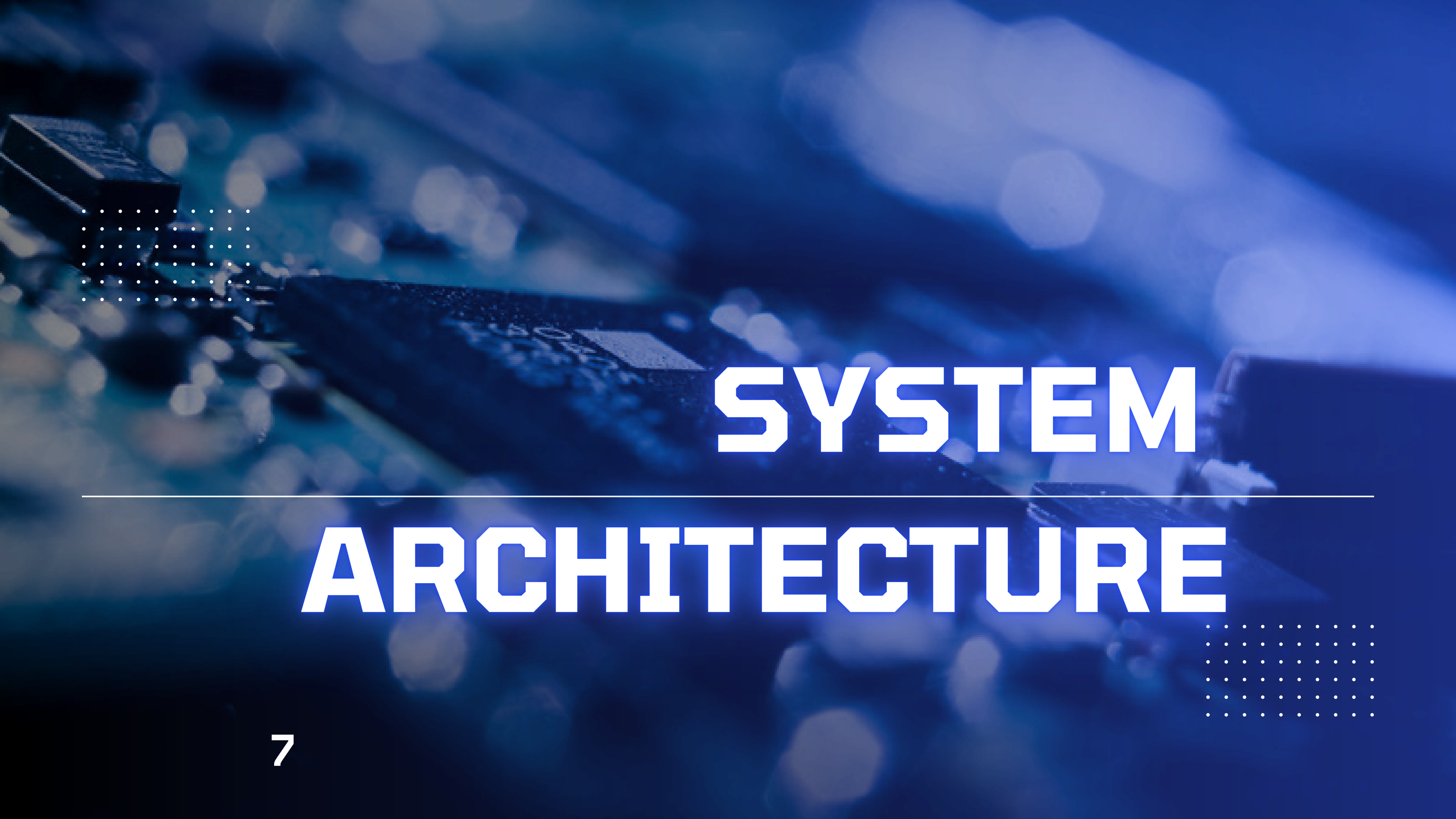
USERS



GNSS # GNSS_R

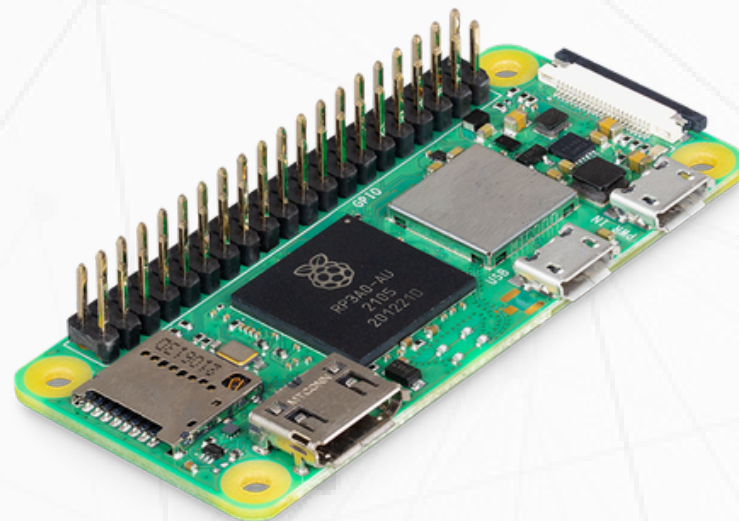
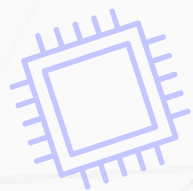




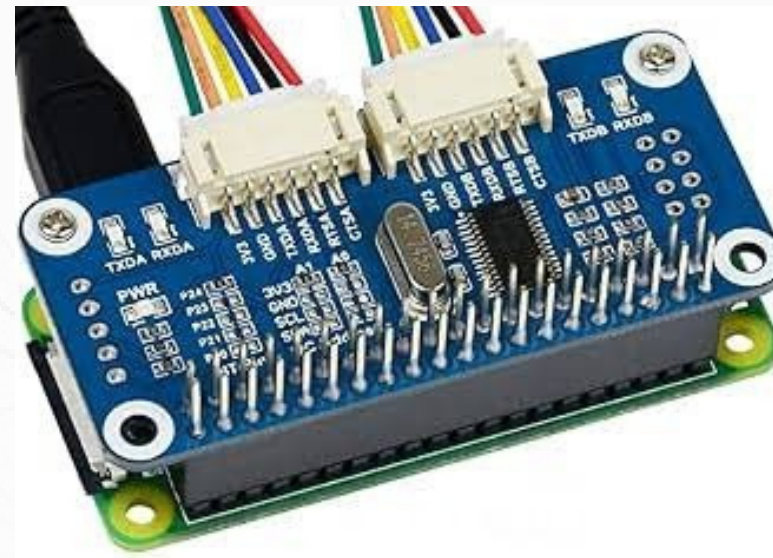


SYSTEM ARCHITECTURE

HARDWARE



Raspberrz PI Zéro 2W



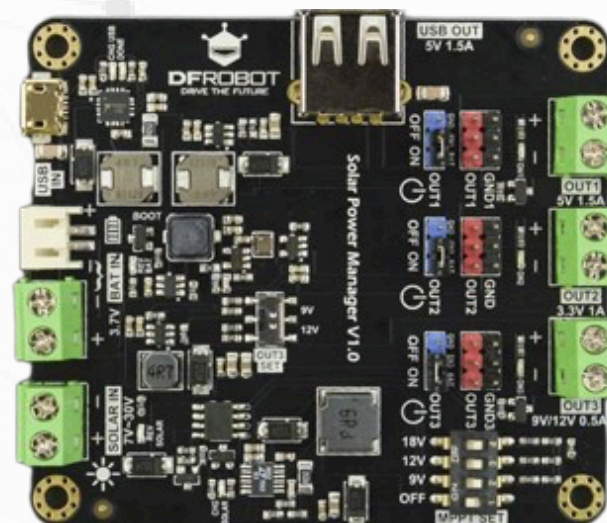
Serial Hat



PX1122_R RECEIVER



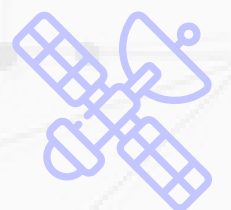
Antenna



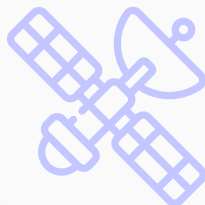
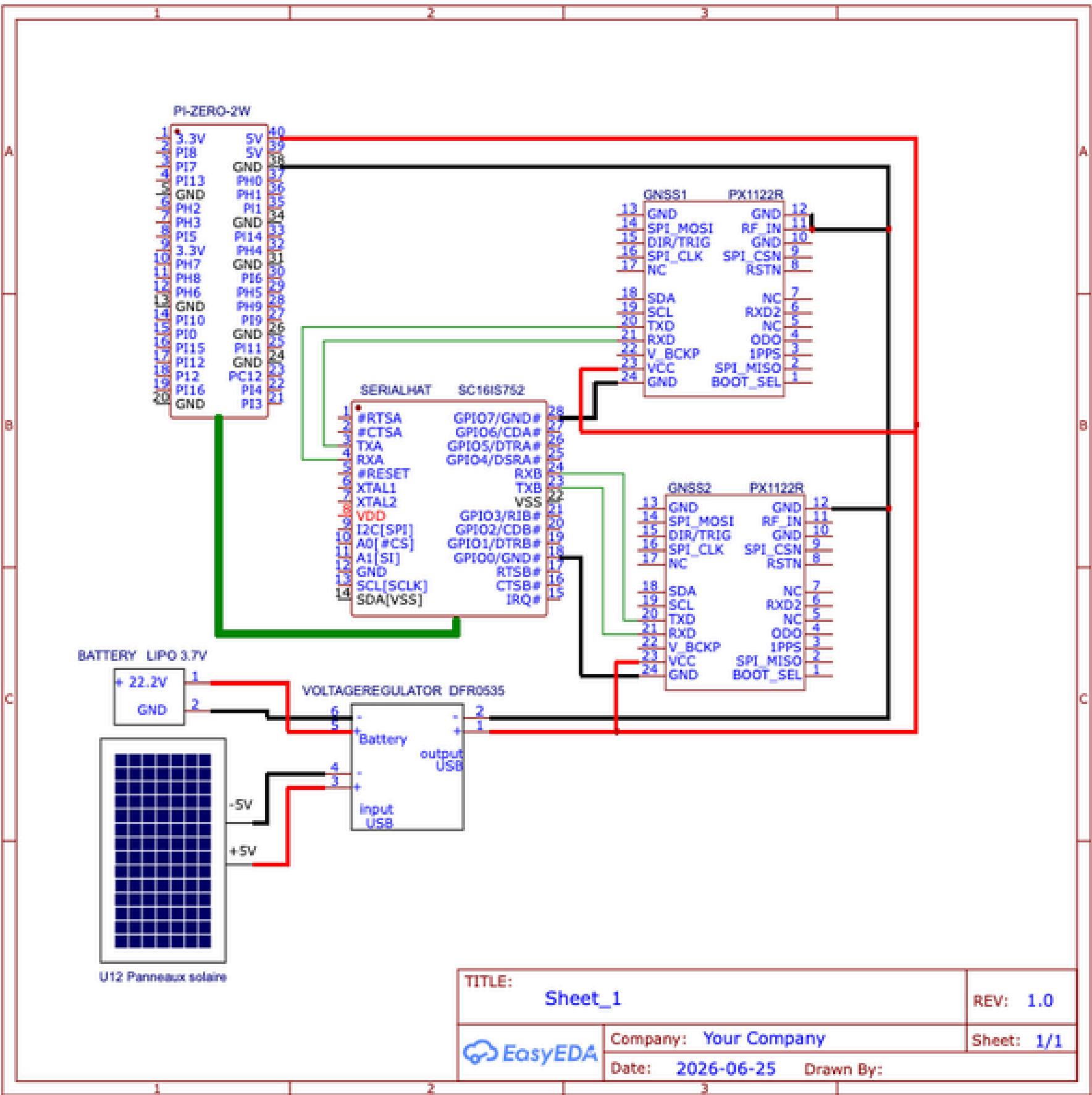
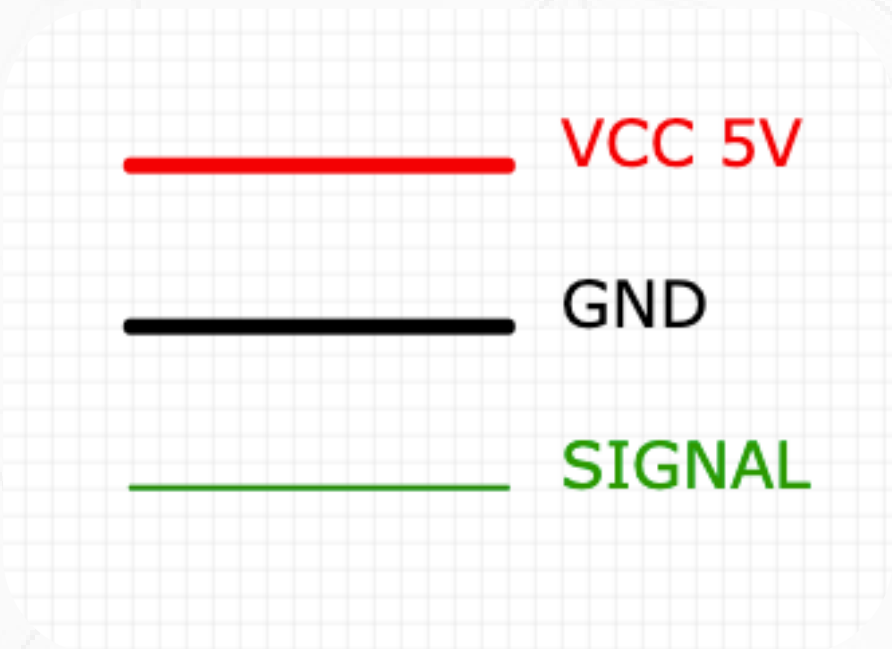
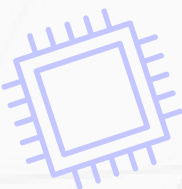
voltage converter

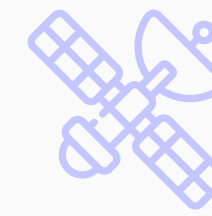
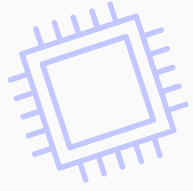


PI LI-ion Battery



System schematic





SOFTWARE



START

Open GNSS1 and GNSS2 serial ports

WHILE (no stop requested):

Create new timestamped .bin files (GNSS1 + GNSS2)

WHILE (segment duration not elapsed):

Read GNSS1 data -> write to .bin file

Read GNSS2 data -> write to .bin file

Close the .bin files

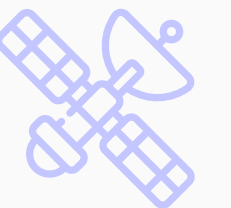
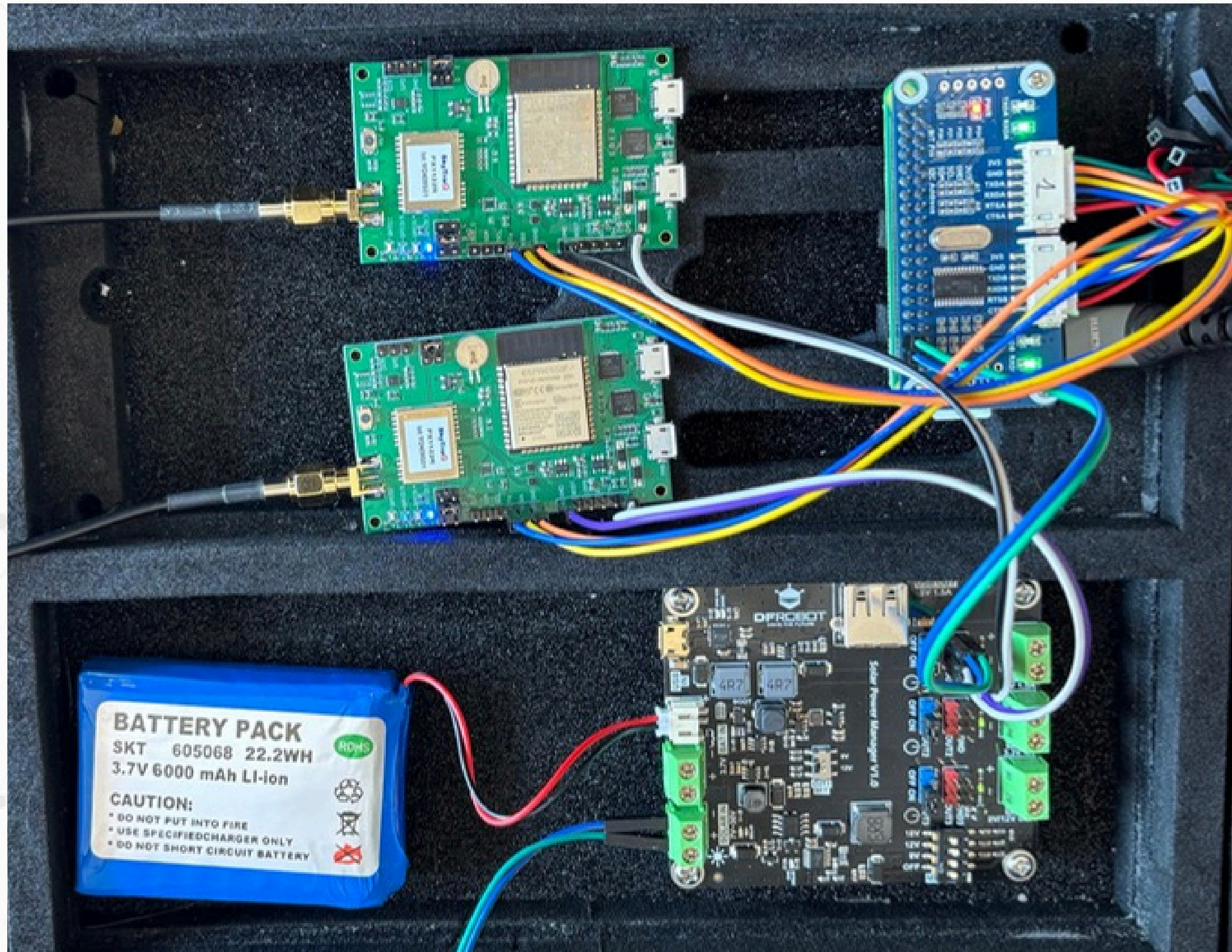
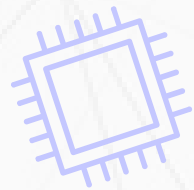
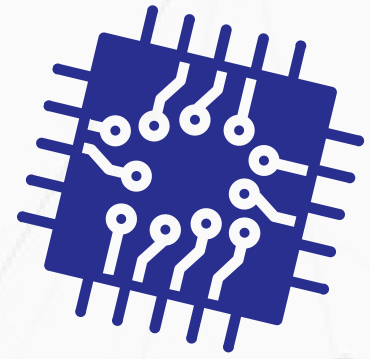
[Background thread] Convert .bin -> .obs + .nav (RINEX)
Save locally

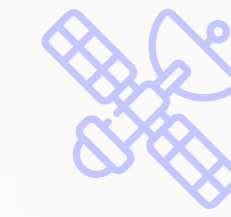
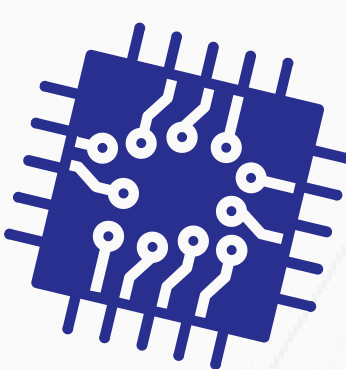
Close serial ports

END

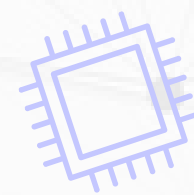


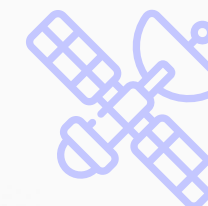
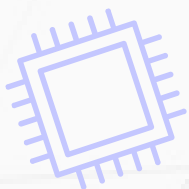
RESULT

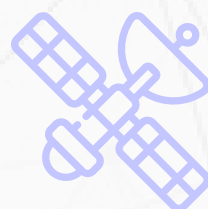
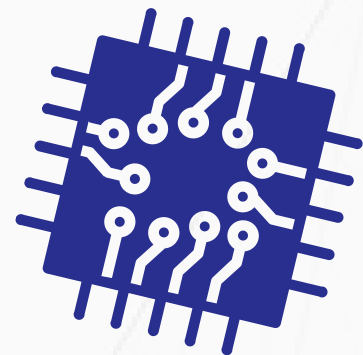




```
admin — pi@pi: ~/Desktop/GNSS/DRONE — ssh pi@pi.local — 139x42
pi@pi:~/Desktop/GNSS/DRONE $ tmux attach -t gnss
[exited]
pi@pi:~/Desktop/GNSS/DRONE $ ls
binary_GNSS1_20260715_154449.bin  convbin  Navig_GNSS1_20260720_162222.nav  Rinex_GNSS2_20260715_154449.obs
binary_GNSS1_20260720_155221.bin  convbin.trace  Navig_GNSS2_20260715_154449.nav  Rinex_GNSS2_20260720_155221.obs
binary_GNSS1_20260720_162222.bin  gnss_alive.txt  Navig_GNSS2_20260720_155221.nav  Rinex_GNSS2_20260720_162222.obs
binary_GNSS2_20260715_154449.bin  gnss_log.txt   Navig_GNSS2_20260720_162222.nav  test_gnss.py
binary_GNSS2_20260720_155221.bin  gnss_status.txt  Rinex_GNSS1_20260715_154449.obs
binary_GNSS2_20260720_162222.bin  Navig_GNSS1_20260715_154449.nav  Rinex_GNSS1_20260720_155221.obs
COMMANDES_TERRAIN.md             Navig_GNSS1_20260720_155221.nav  Rinex_GNSS1_20260720_162222.obs
pi@pi:~/Desktop/GNSS/DRONE $
```



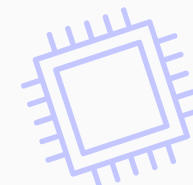


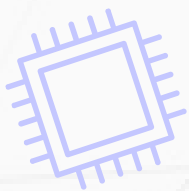
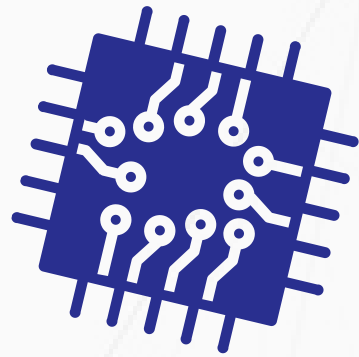


```
admin — pi@pi: ~/Desktop/GNSS/DRONE — ssh pi@pi.local — 137x55

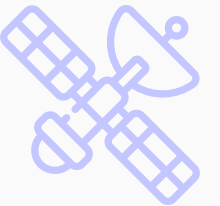
3.04      OBSERVATION DATA      M: Mixed      RINEX VERSION / TYPE
CONVBIN 2.4.2      20260720 223834 UTC PGM / RUN BY / DATE
log: /home/pi/Desktop/GNSS/DRONE/binary_GNSS2_20260720_16222COMMENT
format: SkyTraq      COMMENT
MARKER NAME
MARKER NUMBER
MARKER TYPE
OBSERVER / AGENCY
REC # / TYPE / VERS
ANT # / TYPE
APPROX POSITION XYZ
ANTENNA: DELTA H/E/N
SYS / # / OBS TYPES
SYS / # / OBS TYPES
SYS / # / OBS TYPES
SYS / # / OBS TYPES
SYS / # / OBS TYPES
SYS / # / OBS TYPES
TIME OF FIRST OBS
TIME OF LAST OBS
SYS / PHASE SHIFT
SYS / PHASE SHIFT
SYS / PHASE SHIFT
SYS / PHASE SHIFT
SYS / PHASE SHIFT
SYS / PHASE SHIFT
GLONASS SLOT / FRQ #
GLONASS COD/PHS/BIS
END OF HEADER

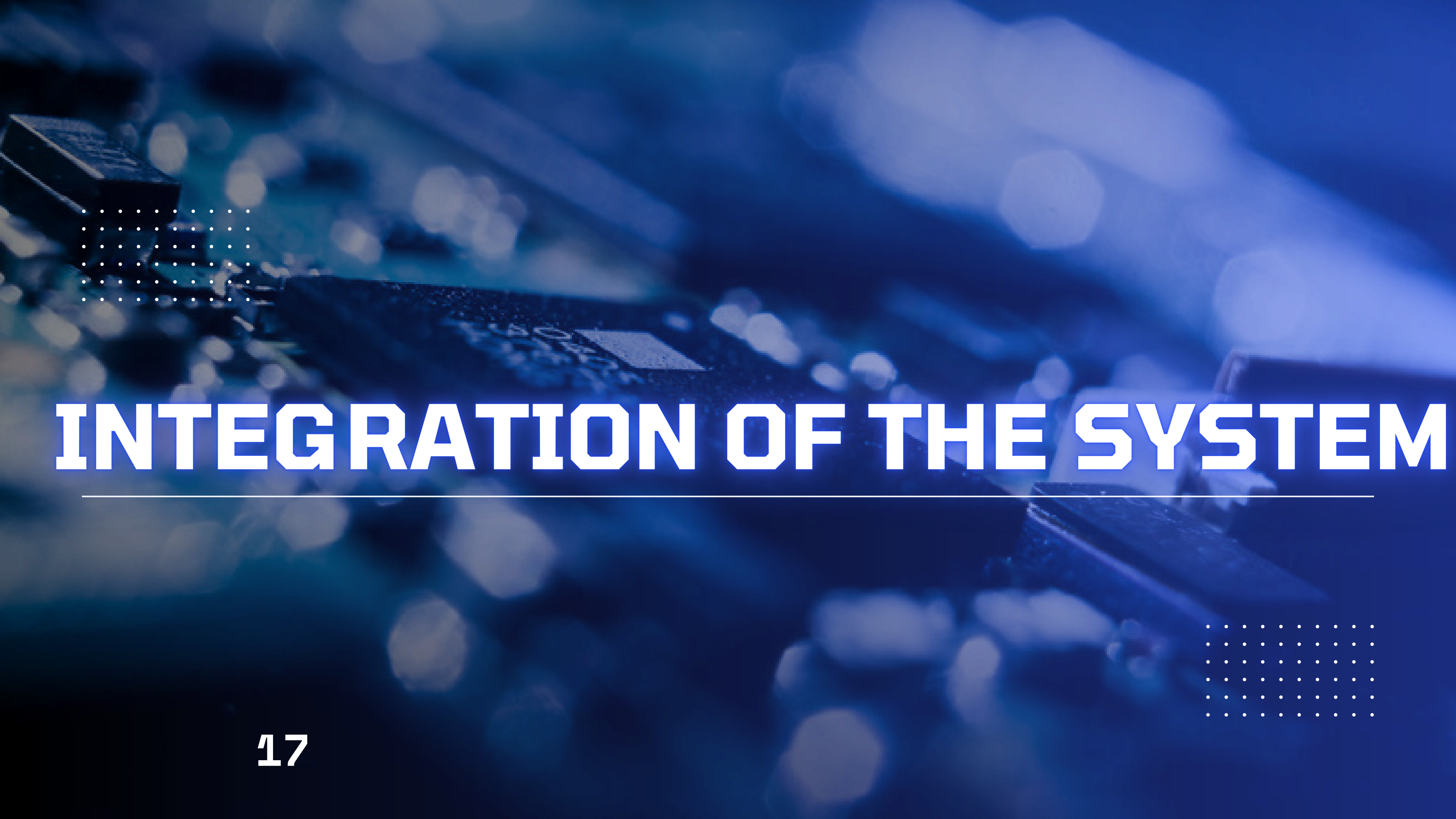
-1517121.4589 -3477662.6491 5110532.7921
0.0000 0.0000 0.0000
G 4 C1C L1C D1C S1C
R 4 C1C L1C D1C S1C
E 4 C1C L1C D1C S1C
J 4 C1C L1C D1C S1C
S 4 C1C L1C D1C S1C
C 4 C1I L1I D1I S1I
2026 7 20 22 22 41.0000000 GPS
2026 7 20 22 38 51.0000000 GPS
G
R
E
J
S
C
0
C1C 0.000 C1P 0.000 C2C 0.000 C2P 0.000
> 2026 7 20 22 22 41.0000000 0 23
G32 21667088.480 -22819890.3781 -2747.709 39.000
G28 20838374.630 -27095397.0341 -146.150 42.000
G31 21278468.919 -32112106.9831 2112.052 47.000
C25 23052388.916 -29399975.014 749.597 45.000
R20 20766865.342 87841631.5251 -593.358 16.000
G25 23086219.031 -29170570.2311 416.130 33.000
R 3 21432643.983 -21787619.183 -3530.185 45.000
R13 21158250.448 -27245960.4011 -1068.633 16.000
R 3 21432651.530 36549850.823 -2745.771 29.000
R19 21809664.982 64424291.549 -4350.466 30.000
C23 25181201.374 53615329.6991 2708.975 31.000
E 7 24202072.401 7313984.4981 -1668.001 28.000
G32 21667092.162 -16725788.0321 -2140.977 28.000
G31 21278463.396 -24508914.5781 1645.811 36.000
E26 25742662.614 64734409.2521 -2715.620 32.000
G25 23086230.479 -22269967.6911 324.240 32.000
R14 21775526.384 80679548.6861 2969.450 29.000
G 2 22813544.318 35618450.9441 -3162.867 34.000
G28 20838367.030 -20582925.3641 -113.802 37.000
R14 21775560.463 62838756.884 2309.713 42.000
C24 25247826.766 72618540.3511 -2443.727 22.000
E 7 24202063.440 -23568485.548 -2176.967 34.000
E26 25742659.019 48345341.352 -3544.185 39.000
> 2026 7 20 22 22 42.0000000 0 23
```





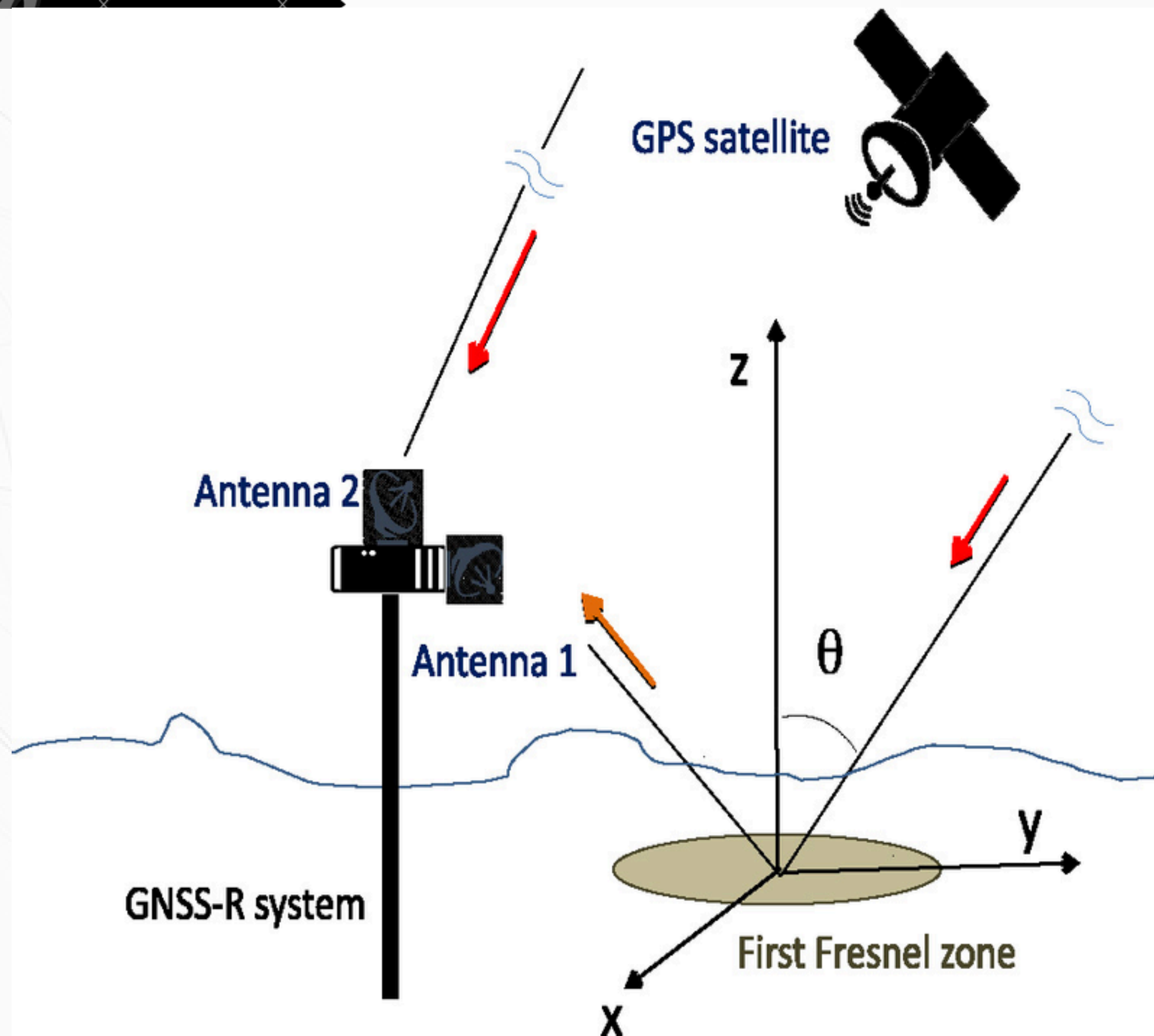
```
admin — pi@pi: ~/Desktop/GNSS/DRONE — ssh pi@pi.local — 137x55
3.04          N: GNSS NAV DATA      M: Mixed      RINEX VERSION / TYPE
CONVBIN 2.4.2          20260720 223834 UTC PGM / RUN BY / DATE
log: /home/pi/Desktop/GNSS/DRONE/binary_GNSS2_20260720_16222COMMENT
format: SkyTraq          COMMENT
                                END OF HEADER
R 3 2026  7 20 22 15  0 .180937349796E-04 .909494701773E-12 .166950000000E+06
    .119424746094E+05 .227031135559E+01 -.931322574615E-09 .000000000000E+00
    -.134273242188E+05 -.995179176331E+00 -.931322574615E-09 .500000000000E+01
    .181179169922E+05 -.224479484558E+01 -.279396772385E-08 .000000000000E+00
R19 2026  7 20 22 15  0 .570509582758E-04 .909494701773E-12 .166950000000E+06
    -.205400976562E+05 -.805911064148E+00 .000000000000E+00 .000000000000E+00
    -.133784428711E+05 -.523270606995E+00 .000000000000E+00 .300000000000E+01
    .693423291016E+04 -.339920330048E+01 .000000000000E+00 .000000000000E+00
C25 2026  7 20 22  0  0 .813732855022E-03 -.903277452835E-11 .000000000000E+00
    .100000000000E+01 .176406250000E+02 .412981488051E-08 -.103400656127E+00
    .847503542900E-06 .576159567572E-03 .103684142232E-04 .528266786194E+04
    .165600000000E+06 -.325962901115E-08 -.204088289128E+01 -.619329512119E-07
    .945316861590E+00 .138187500000E+03 .620750071955E+00 -.686742891312E-08
    -.314298806093E-09 .000000000000E+00 .107200000000E+04 .000000000000E+00
    .240000000000E+01 .000000000000E+00 .400000000000E-09 .400000000000E-09
    .166950000000E+06 .100000000000E+01
G32 2026  7 21  0  0  0 .174568500370E-03 .202362571144E-10 .000000000000E+00
    .830000000000E+02 -.210625000000E+02 .461876381862E-08 -.146652960854E+01
    -.867992639542E-06 .945779099129E-02 .610947608948E-05 .515357058716E+04
    .172800000000E+06 .162050127983E-06 .264250410830E+01 .173225998878E-06
    .969674253042E+00 .264093750000E+03 -.197283062962E+01 -.805247827518E-08
    .282154610015E-09 .100000000000E+01 .242800000000E+04 .000000000000E+00
    .240000000000E+01 .000000000000E+00 .931322574615E-09 .830000000000E+02
    .166986000000E+06 .400000000000E+01
G28 2026  7 21  0  0  0 -.465970486403E-03 .109139364213E-10 .000000000000E+00
    .940000000000E+02 -.304062500000E+02 .473626871317E-08 .209337228664E+01
    -.140443444252E-05 .517320935614E-03 .864453613758E-05 .515362137794E+04
    .172800000000E+06 .106170773506E-06 -.261005354536E+01 .763684511185E-07
    .962243228996E+00 .213031250000E+03 -.311745270882E+00 -.806890753096E-08
    -.196793911542E-09 .100000000000E+01 .242800000000E+04 .000000000000E+00
    .240000000000E+01 .000000000000E+00 -.931322574615E-08 .350000000000E+03
    .166986000000E+06 .400000000000E+01
G31 2026  7 21  0  0  0 -.134117901325E-03 .272848410532E-11 .000000000000E+00
    .440000000000E+02 -.254375000000E+02 .491877631535E-08 .337178054877E+00
    -.111944973469E-05 .109144299058E-01 .863708555698E-05 .515363154411E+04
    .172800000000E+06 -.465661287308E-07 -.256108242263E+01 .968575477600E-07
    .955019772544E+00 .210656250000E+03 .976478249919E+00 -.829891711178E-08
    -.295369446180E-09 .100000000000E+01 .242800000000E+04 .000000000000E+00
    .240000000000E+01 .000000000000E+00 -.135041773319E-07 .440000000000E+02
    .166986000000E+06 .400000000000E+01
G25 2026  7 20 23 59 44 .404865946621E-03 -.318323145621E-11 .000000000000E+00
```



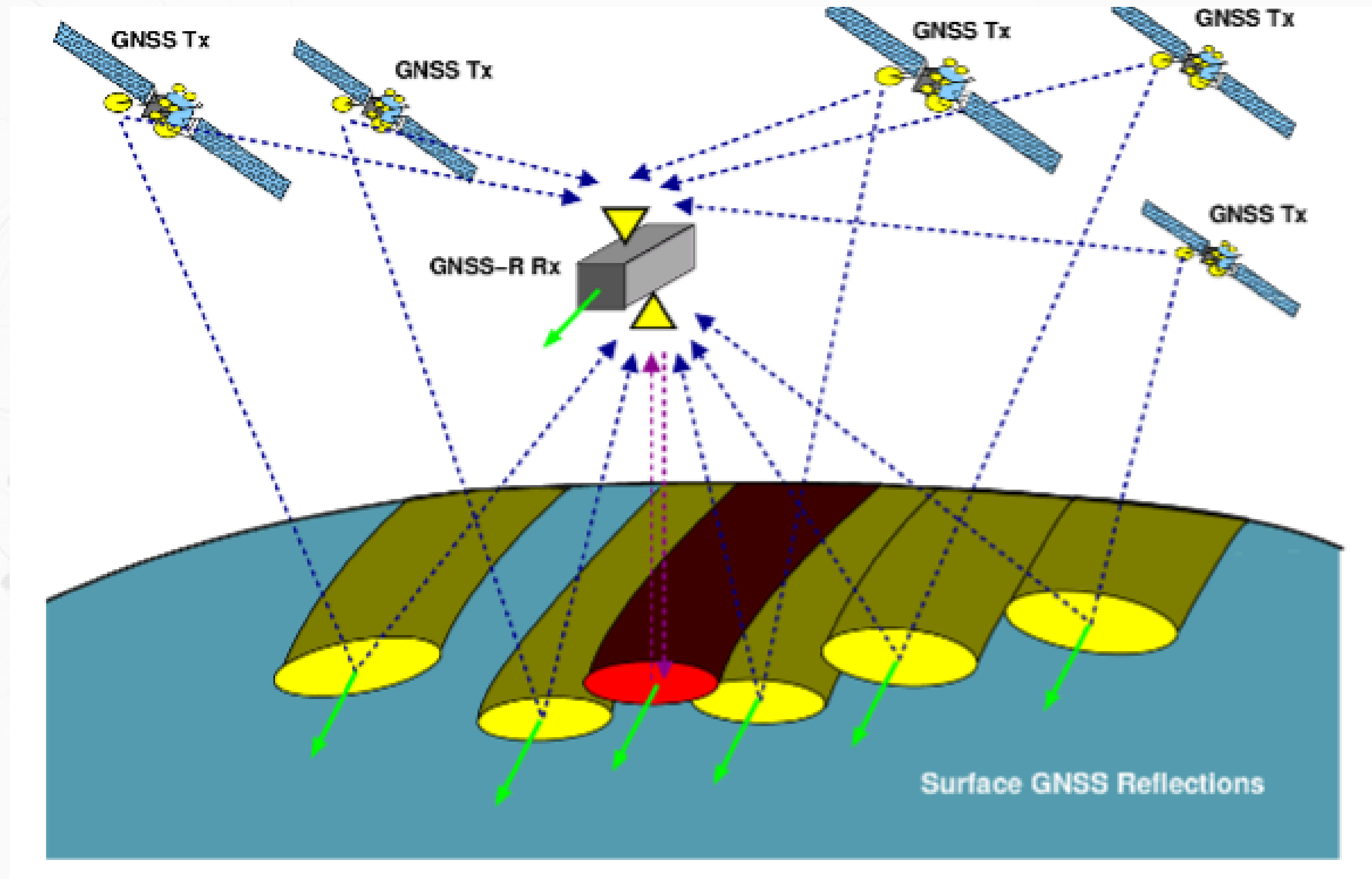


INTEGRATION OF THE SYSTEM

STATIC



MOBILE



CONCLUSION

**THANK
YOU!**

